

Compressible Flow Instabilities in Electronic Liquids

Jack Farrell

Dean Group Meeting

2025-06-03



University of Colorado **Boulder**

Collaborators 😊



Andy Lucas (Boulder)



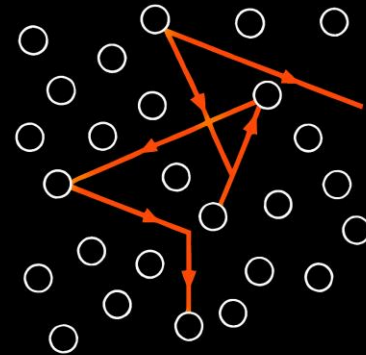
Thomas Scaffidi (Irvine)

Outline

- Nonlinear electron hydrodynamics
- Dyakonov-Shur Instability
 - Fate of growing modes
 - Corbino geometry
 - Ballistic-hydrodynamic crossover
- Idea: accretion shock instability in Corbino
- Conclude

Hydrodynamics as a many-body theory

- In electronic transport, we are dealing with 10^{23} particles (classical or quantum)



- On long-time and length scales, *most functions of these degrees of freedom decay or rapidly oscillate*
- One must identify the *longest-lived* or *slow* modes
- One way to identify these is to study the *conserved* quantities such as (perhaps)

N = number of particles

$\mathbf{P}$ = total momentum

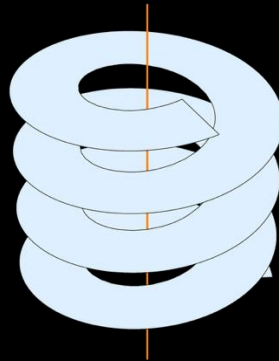
Electron Hydrodynamics

- Consider transport at constant T $v_s^2 \equiv \frac{1}{m} \partial P / \partial n$ Galileo ($\mathbf{u} \rightarrow \mathbf{u} + \mathbf{c}$)
- charge density n , drift velocity $\mathbf{u}$

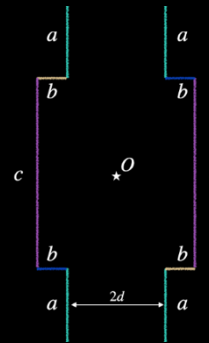
$$mn \partial_t \mathbf{u} = -en \mathbf{E} - \frac{1}{\tau_{\text{mr}}} n \mathbf{u} + \eta \nabla^2 \mathbf{u} - \nabla P(n) + n \mathbf{u} \cdot \nabla \mathbf{u}$$

$$+ \lambda_1 \mathbf{u} \cdot \nabla \mathbf{u} + \lambda_2 \nabla(\mathbf{u}^2) + \lambda_3 \mathbf{u} (\nabla \cdot \mathbf{u}) + O(\mathbf{u}^3)$$

$$+ \text{anisotropic corrections} + \dots$$

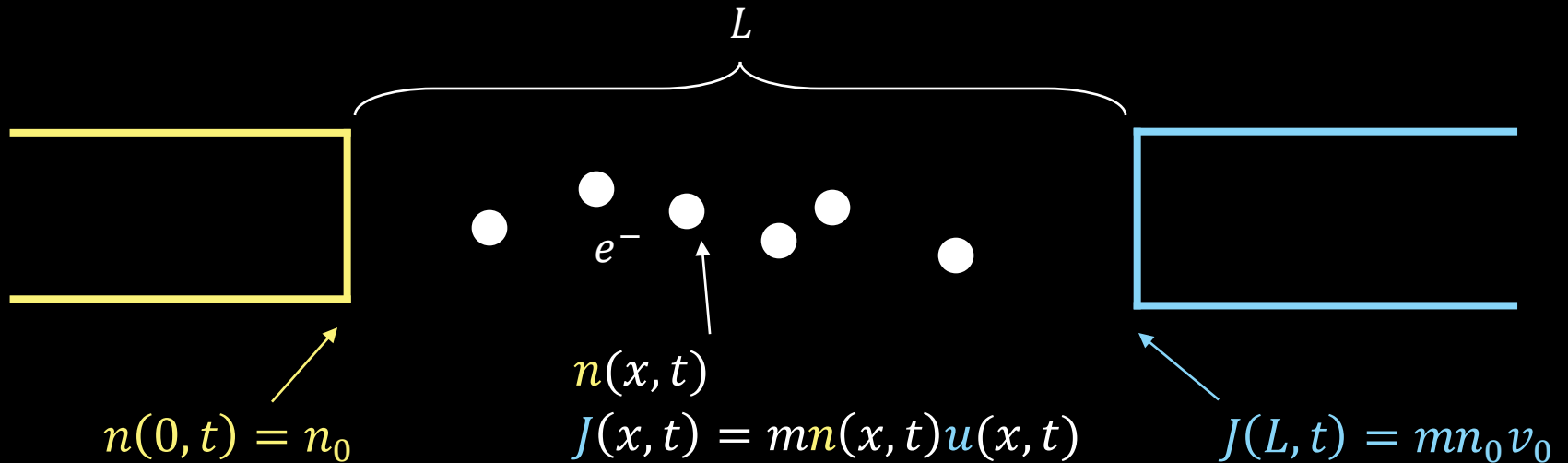


[JF, Huang, Lucas, 2022]



[JF, Hicks, 2025 (to appear)]

Dyakonov-Shur Instability

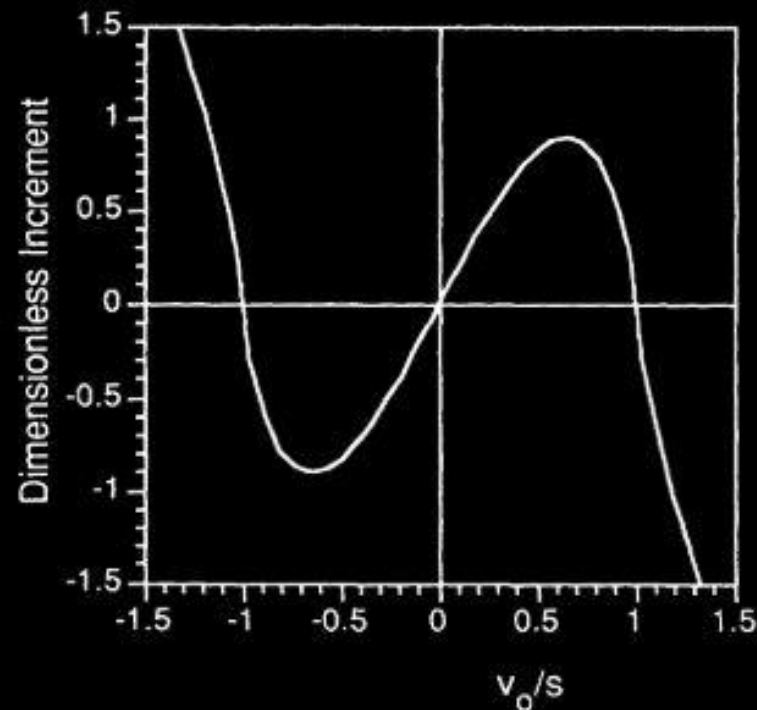


- [Dyakonov, Shur, 1993] :
 - normal modes $\delta n, \delta v \sim e^{ikx - i\omega t}$
 - $\text{Re}(\omega) = \frac{\pi v_s}{2L}$ —Terahertz!
 - $\text{Im}(\omega) = \frac{v_s^2 - v_0^2}{2Ls} \log \left| \frac{v_s + v_0}{v_s - v_0} \right| > 0$ if $v_0 < v_s$

Dyakonov-Shur Instability

[DS 1993]

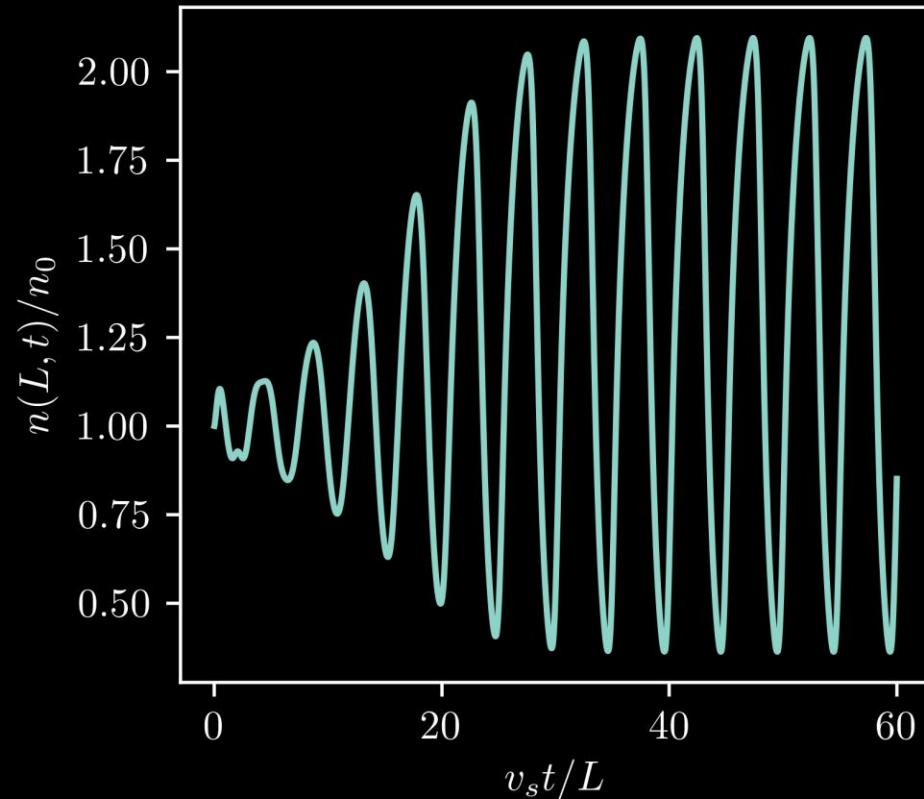
$$\omega = \frac{|v_s^2 - v_0^2|}{2Lv_s} + i \left(\frac{v_s^2 - v_0^2}{2Lv_s} \log \left| \frac{v_s + v_0}{v_s - v_0} \right| \right)$$



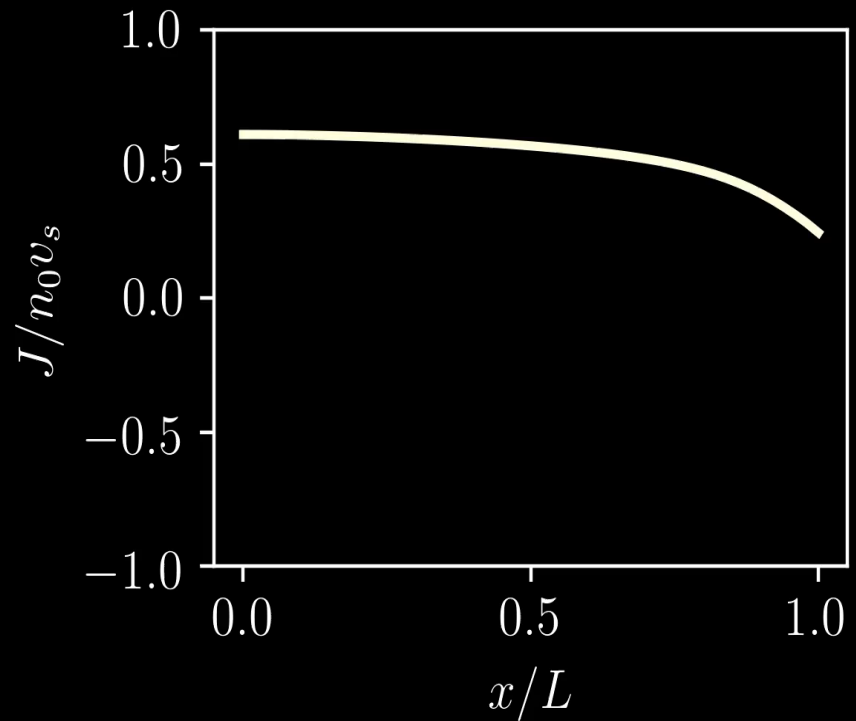
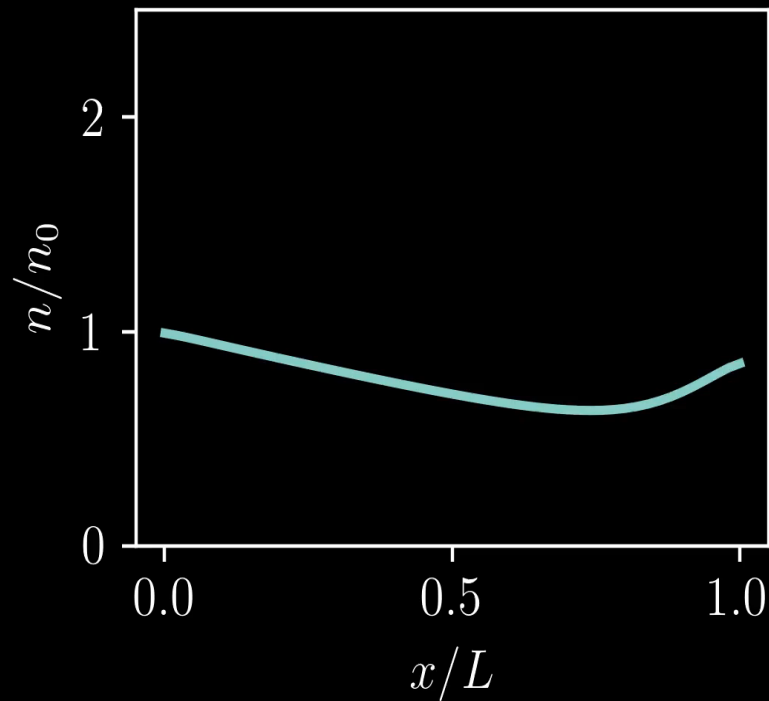
Nonlinear Simulations

Instability
terminates in an
oscillator

[Mendl, Polini, Lucas, 2019]

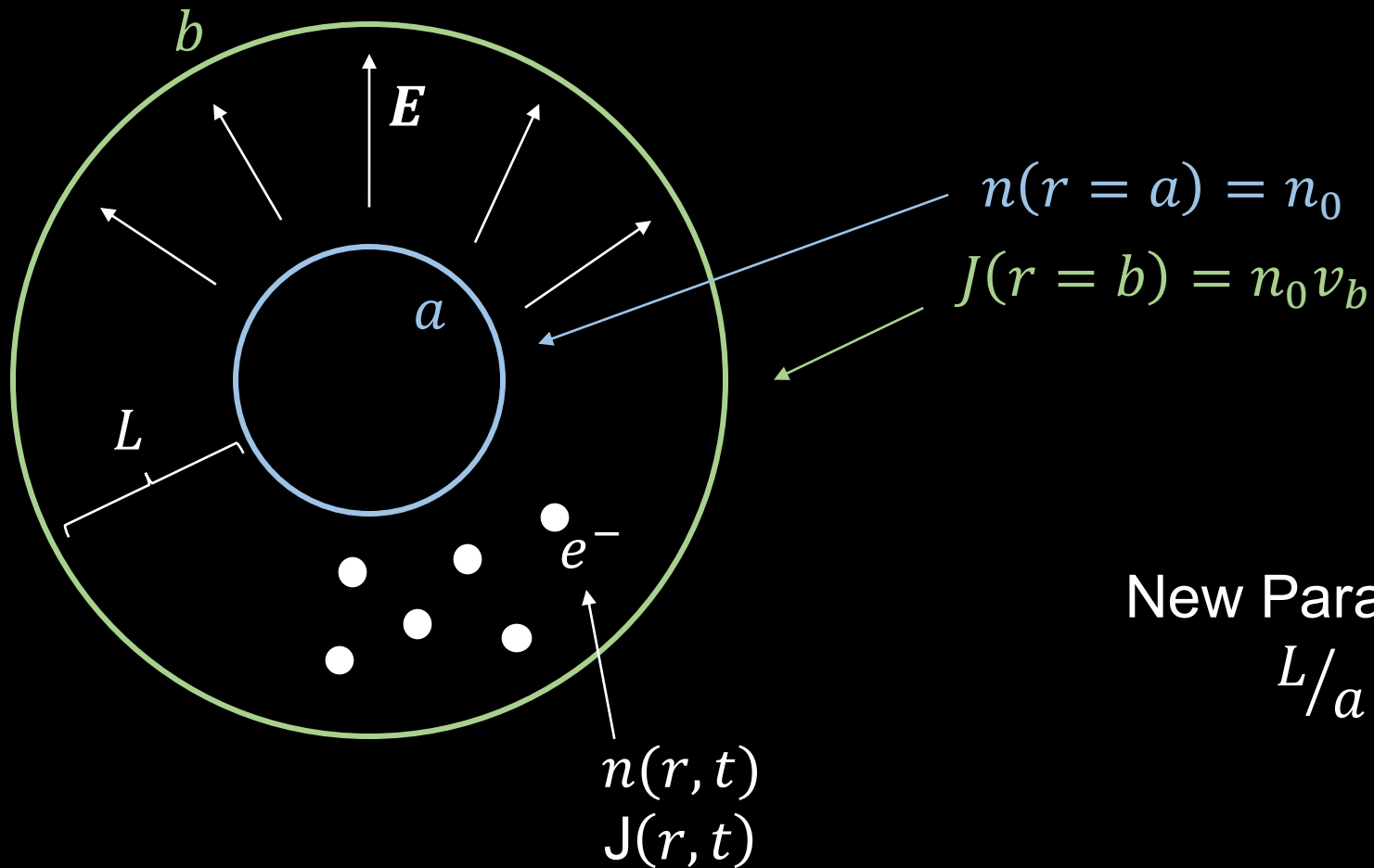


Nonlinear Simulations



DS in a Corbino Disk

[JF, Grisouard, Scaffidi, 2022]



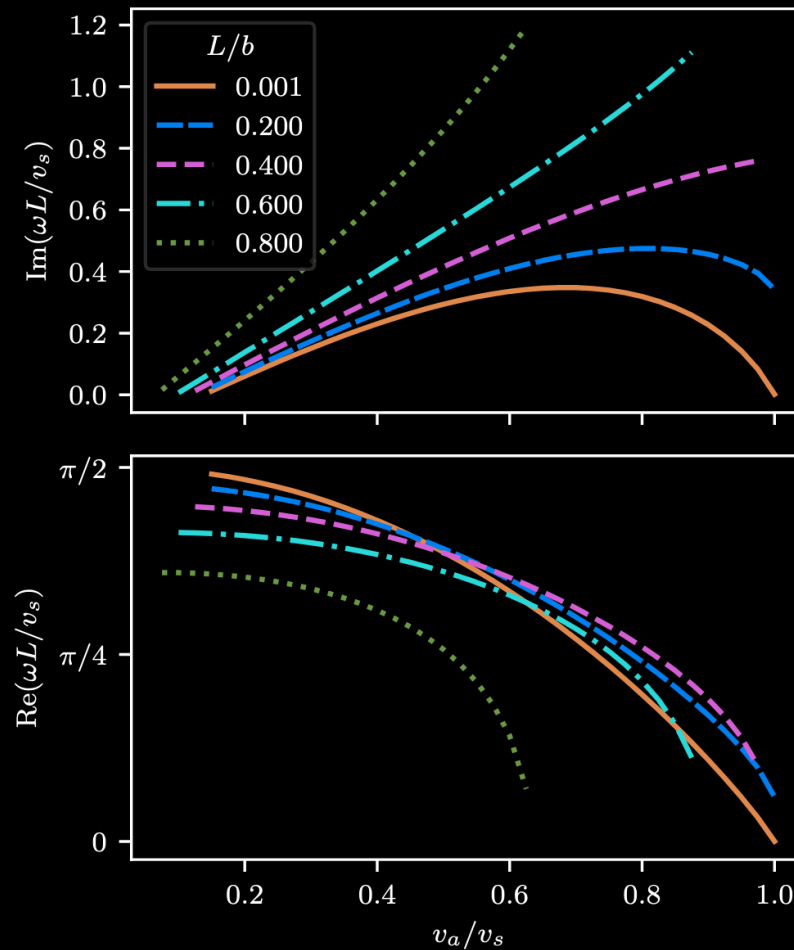
Navier Stokes in Polar Coordinates

$$\begin{aligned}
 & \underbrace{\frac{\partial J}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{J^2}{n} \right)}_{\text{Advection}} + \underbrace{v_s^2 \frac{\partial n}{\partial r}}_{\text{Pressure}} - \overbrace{\eta \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \left(\frac{J}{n} \right) \right) - \frac{1}{r^2} \frac{J}{n} \right)}^{\text{Viscous Diffusion}} \\
 & \qquad \qquad \qquad = \underbrace{\left(\frac{\gamma n v_b b}{r} - \frac{n v_b^2 b^2}{r^3} - \gamma J \right)}_{\text{Momentum Relaxation \& driving}}
 \end{aligned}$$

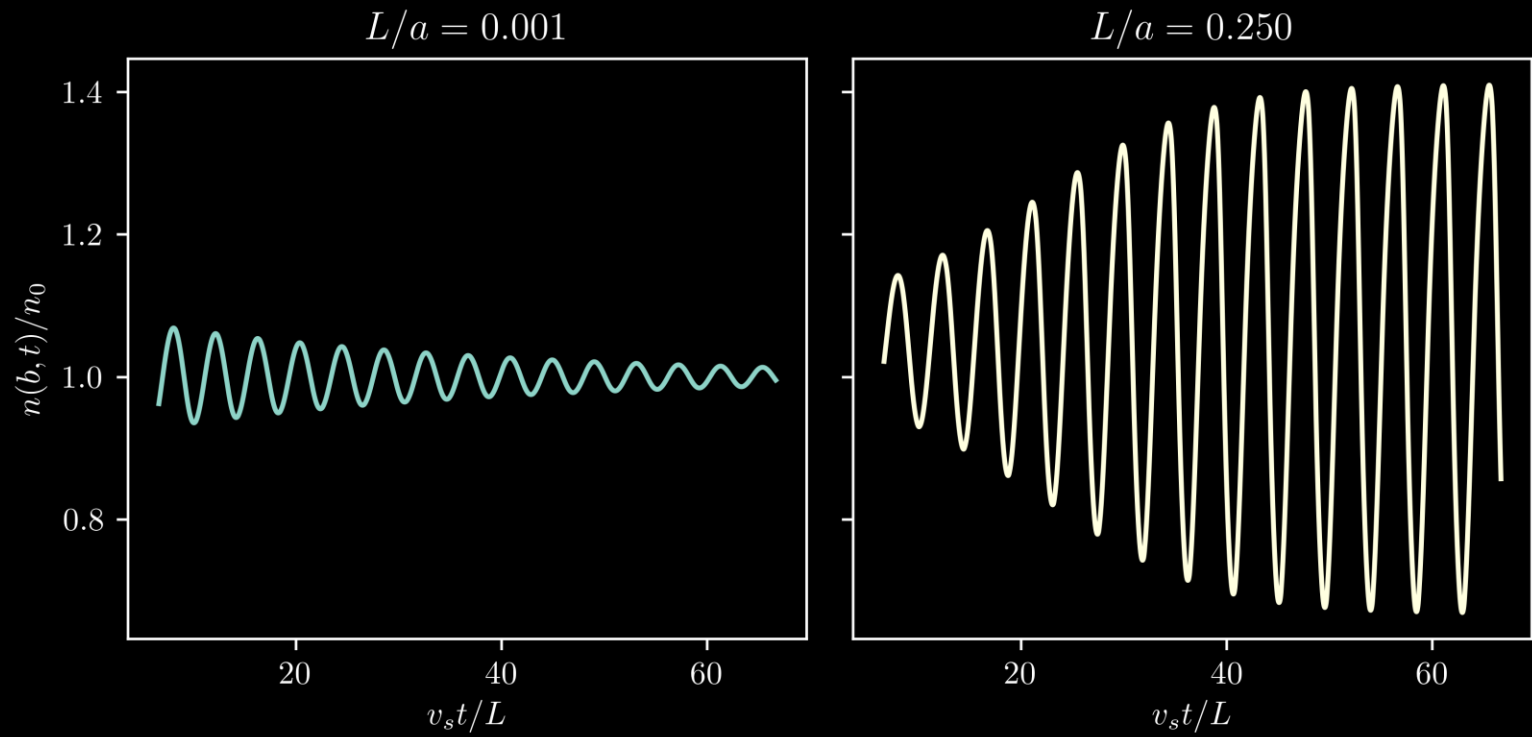
$$\frac{\partial n}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (rJ) = 0$$

Linear Response

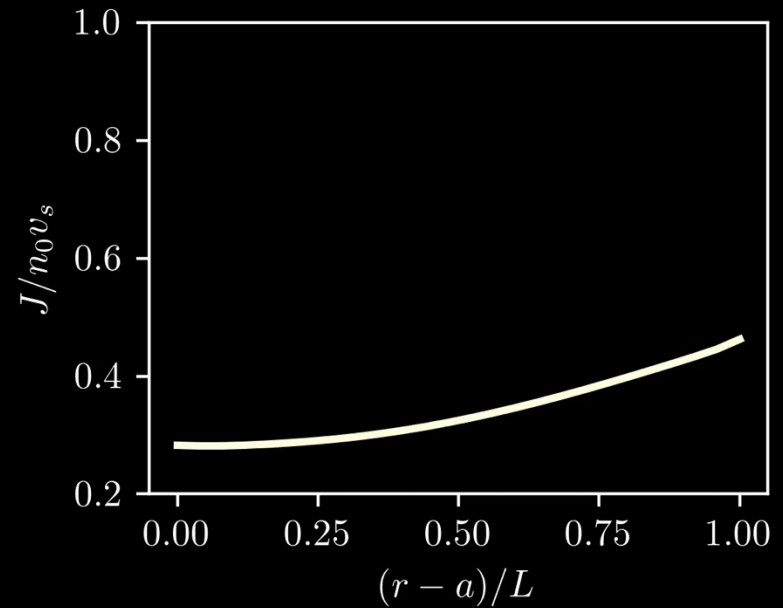
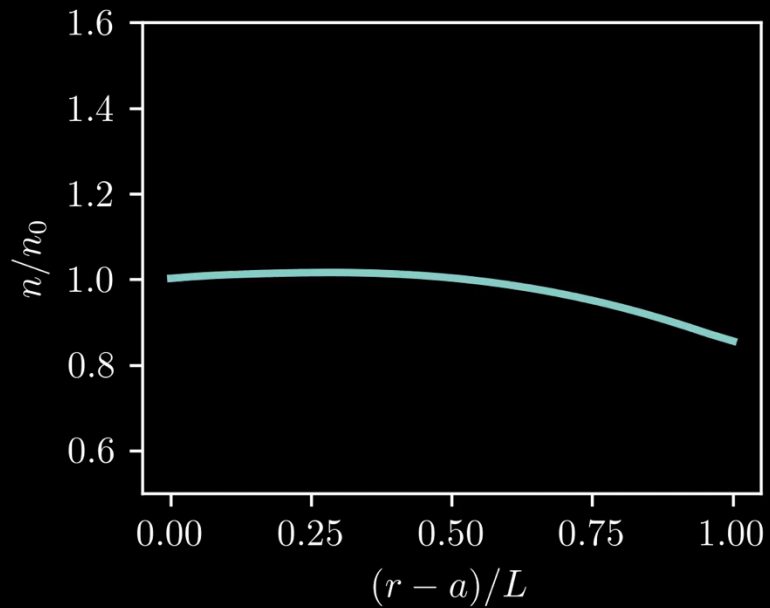
- Corbino geometry enhances the instability



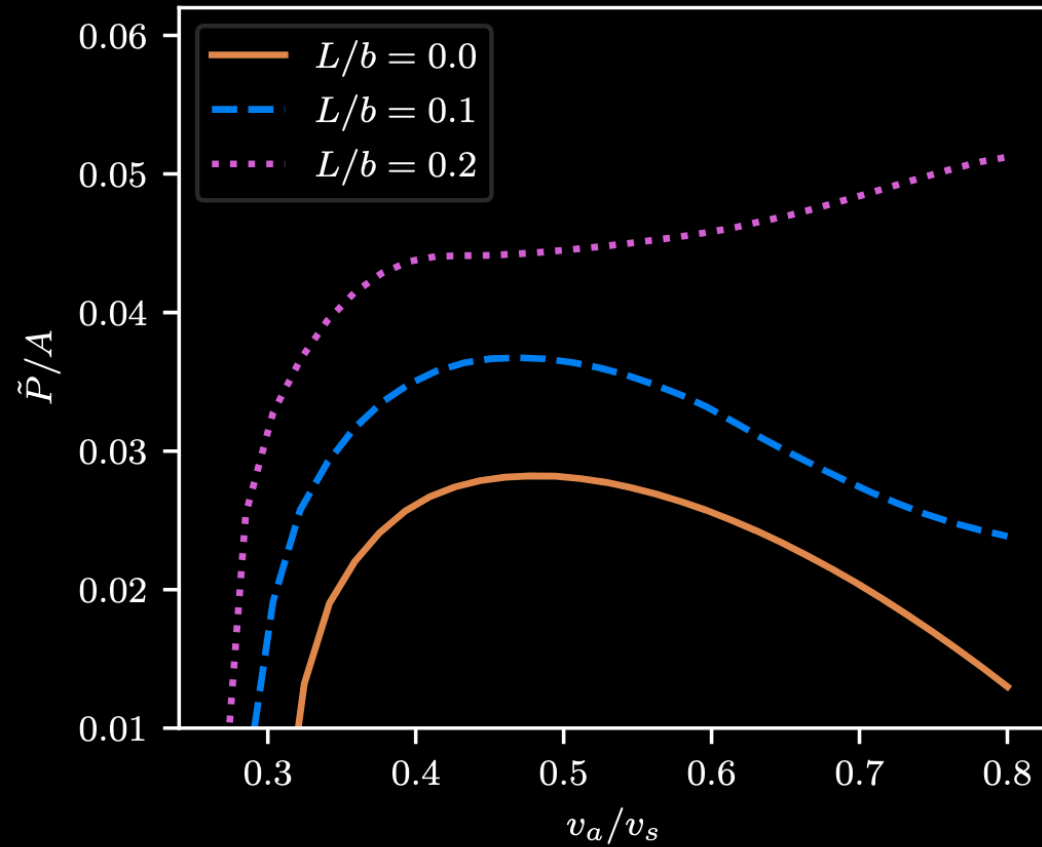
Nonlinear Simulations



Nonlinear Simulations



Nonlinear Simulations



Symmetric BCs work too?

APPLIED PHYSICS LETTERS **97**, 263504 (2010)

Plasma oscillations and terahertz instability in field-effect transistors with Corbino geometry

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(Received 30 September 2010; accepted 7 December 2010; published online 29 December 2010)

Propagating between the contacts of a field-effect transistor (FET), plasma waves in its channel can become unstable and lead to generation of terahertz radiation. While previous studies of this instability concentrated on rectangular FETs, alternative geometries present fresh opportunities. We studied theoretically plasma oscillations in a gated FET with Corbino geometry where, in contrast with the rectangular FET, the oscillations become unstable at symmetric boundary conditions. Moreover, their lowest eigenfrequency is almost twice as high as that in the rectangular FET at comparable instability increments. These advantages make the Corbino FET promising for practical realizations of terahertz oscillators. © 2010 American Institute of Physics. [doi:[10.1063/1.3532850](https://doi.org/10.1063/1.3532850)]

Obstacles

- Boundary conditions (in particular $J(L, t) = \text{const.}$)
- TeraHertz antennae... without this, Power $\sim \text{pW}$
- Momentum relaxation, $\text{Im}(\omega) < 0$ when $\gamma_{\text{mr}} > v_s/L$ [Mendl, Polini, Lucas, 2022]
- ~~Crossover to ballistic physics~~

Punchline: ballistic transport should favour the DS instability; in particular, Power $\sim \tau_{ee}^2$ near instability onset

A nonlinear cartoon

- Take gas of fermions with $\varepsilon(\mathbf{p}) = \mathbf{p}^2/2m$
- They have a distribution function $f(\mathbf{x}, \mathbf{p}, t)$, which in equilibrium and near $T = 0$ is $f_0 = \Theta(\epsilon_F - p^2/2m)$

$$\partial_t f + \frac{\mathbf{p}}{m} \cdot \nabla f = \mathcal{C}[f]$$

Hydrodynamic limit:

Local thermodynamic equilibrium
consistent with conserved *particle
number* and *momentum*

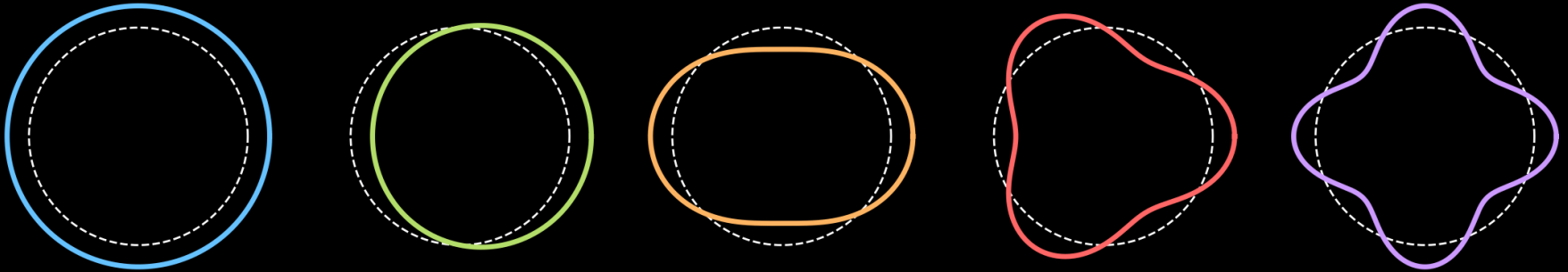
$$f(\mathbf{x}, \mathbf{p}, t) = \Theta\left(\mu(\mathbf{x}) - \mathbf{p} \cdot \mathbf{u}(\mathbf{x}) - \frac{\mathbf{p}^2}{2m}\right)$$

Ballistic limit:

$$\mathcal{C}[f] = 0$$

$$\mathcal{C}[f] \approx \frac{1}{\tau_{ee}(\mathbf{p})} \left\{ \Theta\left(\mu(\mathbf{x}) - \mathbf{p} \cdot \mathbf{u}(\mathbf{x}) - \frac{\mathbf{p}^2}{2m}\right) - f \right\}$$

A nonlinear cartoon



$$f - f_0 \equiv \delta(\epsilon_F - \varepsilon) \times \sum_n a_n(x, t) e^{in\theta} \quad \left\{ \begin{array}{l} n(x, t) = n_0 + \frac{m}{2\pi} a_0(x, t) \\ J(x, t) = \frac{mv_F}{2\pi} a_1(x, t) \end{array} \right.$$

- Giving

$$\partial_t a_n + \frac{v_F}{2} \partial_x (a_{n+1} + a_{n-1}) = c_n$$

- Final ingredient: presence of background flow u_0

$$\partial_t \rightarrow \partial_t + u_0 \partial_x$$

A nonlinear cartoon

$$\partial_t a_0 + u_0 \partial_x a_0 + v_F \partial_x a_1 = 0$$

$$\partial_t a_1 + u_0 \partial_x a_1 + \frac{v_F}{2} \partial_x (a_2 + a_0) = 0$$

$$\partial_t a_2 + u_0 \partial_x a_2 + \frac{v_F}{2} \partial_x (a_3 + a_1) = 0 - \frac{1}{\tau_{ee}} a_2 - \frac{1}{\tau_{ee}} \left(\frac{a_0^2}{m v_F^2} + \frac{4 a_1^2}{m v_F^2} \right)$$

⋮

$$\partial_t a_n + u_0 \partial_x a_n + \frac{v_F}{2} \partial_x (a_{n+1} + a_{n-1}) = 0 - \frac{1}{\tau_{ee}} a_n$$

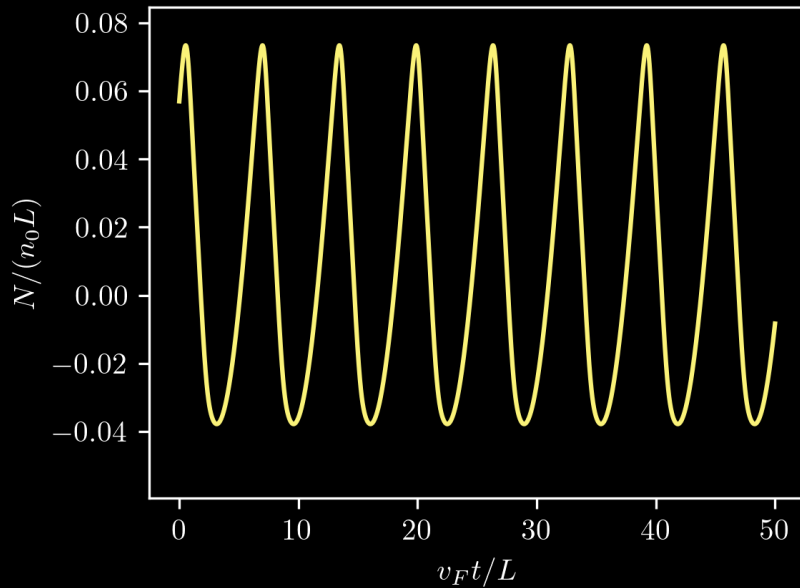
at lowest order in τ_{ee} , right hand side vanishes, and one can show:

$$\left. \begin{aligned} \partial_t n + \frac{1}{m} \partial_x J &= 0 \\ \partial_t J + \partial_x \left(\frac{J^2}{m n_0} \right) + \partial_x \left((2\pi \hbar^2) \frac{n^2}{4\pi} \right) &= 0 \end{aligned} \right\} \begin{array}{l} \text{Hydrodynamics in the} \\ \text{"Boussinesq"} \\ \text{approximation} \end{array}$$

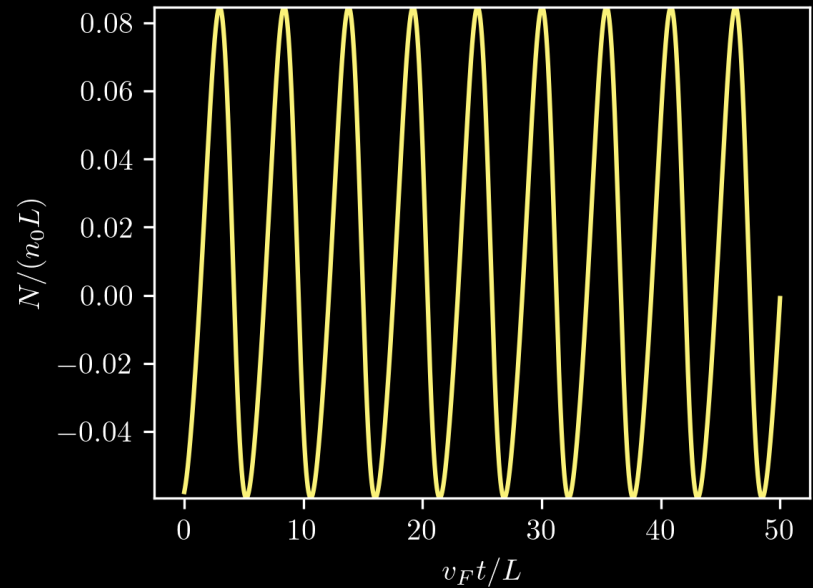
Numerics

- Truncate at $n_{\max} = 40$

$$\frac{\tau_{ee} v_F}{L} = 0.25$$

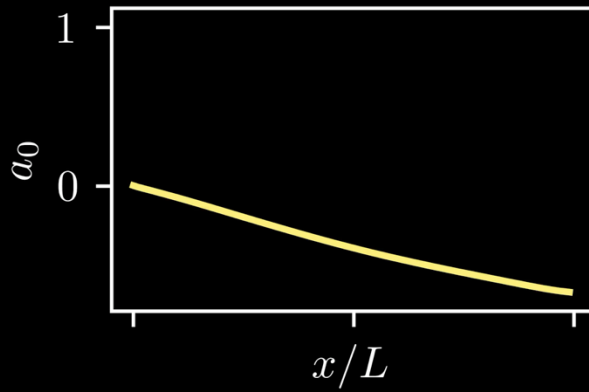


$$\frac{\tau_{ee} v_F}{L} = 1.25$$



Numerics

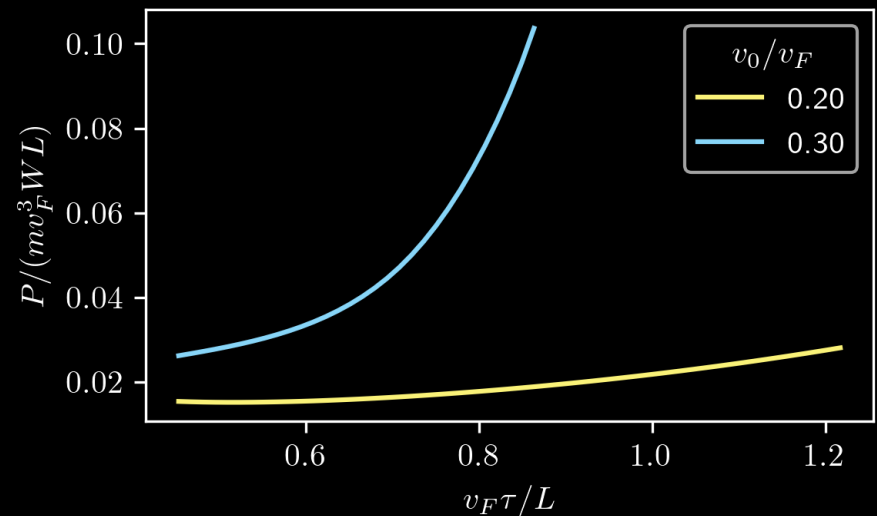
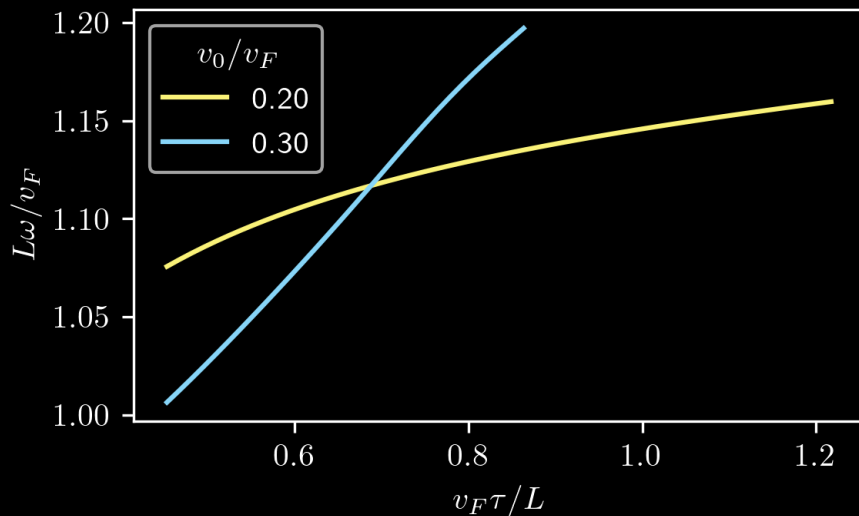
$$\frac{\tau_{ee} v_F}{L} = 1.25$$



Characterization

- Imagine the device is separated from a metallic gate by an oxide, forming a capacitor
- Power from charge-discharge cycles:

$$\text{Power} \approx \frac{1}{T} \left| \int_0^T \frac{dt}{T} \frac{(eN)^2}{2C} \cos(2\pi t/T) \right|$$



Other flow instabilities?

- Shallow water analogies were / are popular in astrophysics

A Shallow Water Analogue of the Standing Accretion Shock Instability: Experimental Demonstration and Two-Dimensional Model

Thierry Foglizzo,¹ Frédéric Masset,^{2,1} Jérôme Guilet,^{3,1} and Gilles Durand¹

¹*AIM, IRFU/SAP, CEA Saclay, 91191, France*

²*ICF, UNAM, 62210, Mexico*

³*DAMTP Cambridge, CB3 0WA, UK*

(Dated: accepted December 13, 2011)

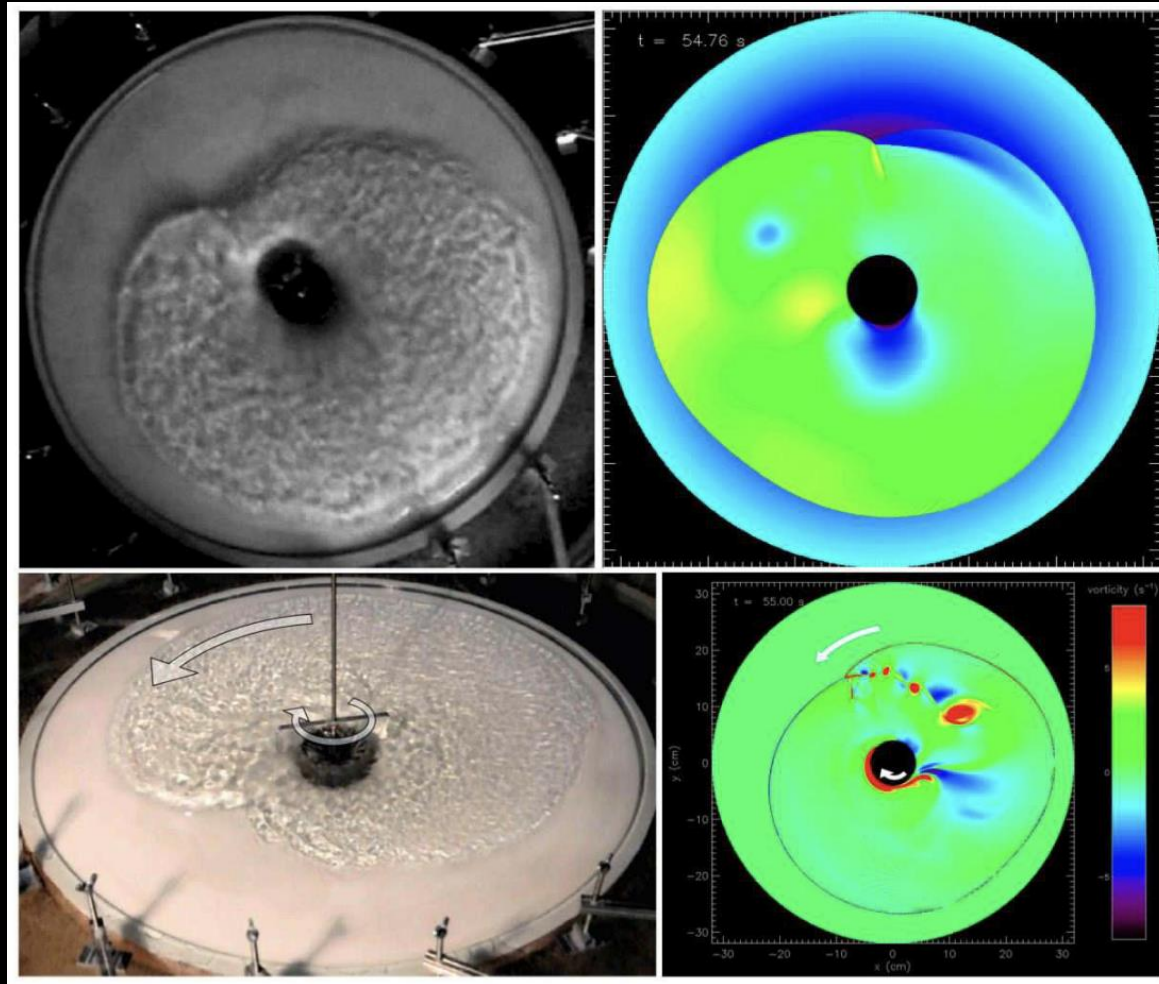
Despite the sphericity of the collapsing stellar core, the birth conditions of neutron stars can be highly non spherical due to a hydrodynamical instability of the shocked accretion flow. Here we report the first laboratory experiment of a shallow water analogue, based on the physics of hydraulic jumps. Both the experiment and its shallow water modeling demonstrate a robust linear instability and nonlinear properties of symmetry breaking, in a system which is one million times smaller and about hundred times slower than its astrophysical analogue.

PACS numbers: 97.60.Bw, 95.30.Lz, 47.20.-k, 01.50.Pa



Half-baked idea:

- Angular perturbations cause the hydraulic jump to “shake”



Thanks for your time! 😊

- Dyakonov-Shur terminates as a coherent nonlinear oscillator
 - In Corbino, bias velocity can be lower, power higher
 - Quasi-ballistic transport is not a concern
-
- Still, there is a need for genuinely new indirect observables of DS
 - *e.g.* local heating?

Hydrodynamics with Helical Symmetry

[JF, Huang, Lucas, 2022]

- Only a combination of rotation and translation is a symmetry: $K_z = P_z + L_z/\xi$
- Equation of motion in-plane velocity v_A is written in terms of a stress tensor τ

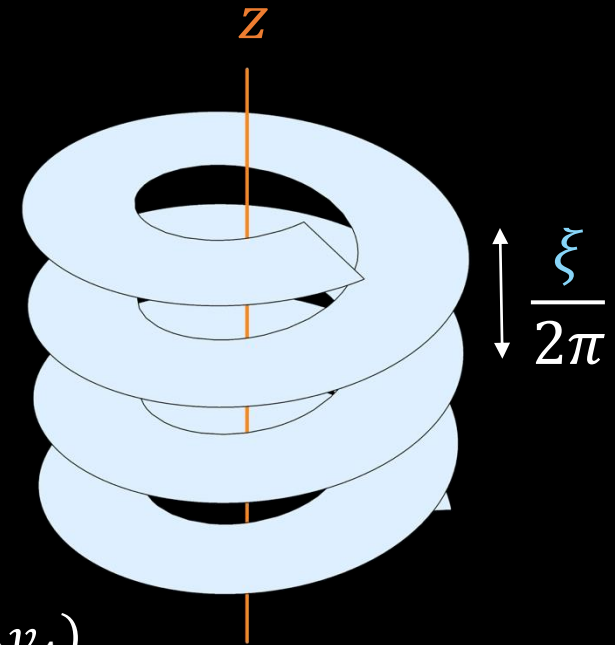
$$\rho_0 \partial_t v_A = -\partial_i \tau_{iA}$$

- In-plane stress tensor:

$$\tau_{AB} = m c_s^2 n \delta_{AB} + m \alpha \epsilon_{AB} \partial_z n + \eta (\partial_A v_B + \partial_B v_A) \dots$$

- Novel *dissipationless* transport coefficient α has a value fixed by symmetry alone:

$$\alpha = \frac{n_0 \xi}{2}$$



- Imagine applying an electric field $\mathbf{E} = E \hat{z}$; a submerged object with area A feels a torque Γ_z :

$$\Gamma_z = 2\alpha(eEA)$$

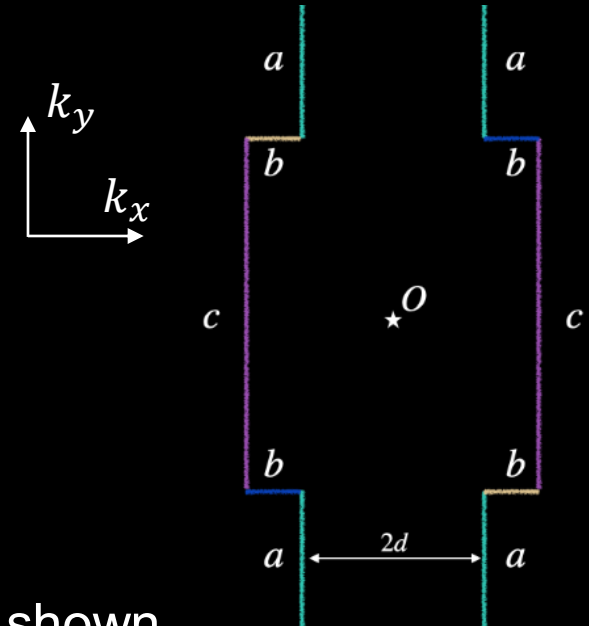
Hydrodynamics with an open Fermi Surface

[JF, Hicks, Lucas, TBD]

- Fermi surface wraps the Brillouin zone
- Low energy scattering events can change k_y by a reciprocal lattice vector (Umklapp)

$$k_y \rightarrow k_y + G_y$$

- only $n(x, y, t)$, $v_x(x, y, t)$ are hydrodynamic
- checked in kinetic theory for the cartoon FS shown



$$\partial_t n = \rho \partial_x v_x + D_{xx} \partial_x^2 n + D_{yy} \partial_y^2 n$$

$$\rho_0 \partial_t v_x = m c_s^2 \partial_x n + \eta_{xx} \partial_x^2 v_x + \eta_{yy} \partial_y^2 v_x$$

- Resistance sensitive to relative angle between channel and FS, and if channel is in y direction, purely from D_{yy} “incoherent conductivity”!